

- Q.27 Writes steps for selection of wires and cables.
- Q.28 What is Earthing?
- Q.29 Compare the different wiring system.
- Q.30 Write short note on MCB.
- Q.31 Write short note on Market survey.
- Q.32 Discuss different types starter.
- Q.33 Compare indoor and outdoor substation.
- Q.34 Estimate and costing of 3 phase motor load.
- Q.35 Explain about mounting arrangements positioning of switch board.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Discuss the classification of installation diagram and explain in briefly about each.
- Q.37 Explain key diagram of 66kv/11kv substation.
- Q.38 Explain steps of preparation of tender documents.

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**3rd Sem / Electrical Engineering, Power Station.,
Elect & Eltx. Engg.**

Sub : Estimating & Costing in Electrical Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the maximum number of lighting points that can be connected in a circuit?
- a) 5 b) 10
c) 8 d) 12
- Q.2 What is the range of current rating of single pole MCBs commercially available?
- a) 2.0 to 6 amps b) 0.5 to 60. Amps
c) 1.0 to 62 amps d) 6.0 to 32 amps
- Q.3 Use of ELCB is advisable for
- a) Human safety
b) Ease to reset when trips
c) Protecting from short circuit
d) None
- Q.4 What is the maximum load that can be connected in a circuit connecting only lighting points?
- a) 500 watts b) 750 watts
c) 800 watts d) 1000 watts

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- Q.5 Which set of rules are to be verified on completion of wiring on any new installation?
- a) IE rules, 1950 b) IE rules, 1956
c) IE rules, 1960 d) None of these
- Q.6 Which among these is a methods of wiring?
- a) Joint box b) Tee system
c) Loop is system d) All of these
- Q.7 Which document acts as an invitation to contractors to bid?
- a) Letter of Acceptance
b) Tender Notice or NIT
c) Purchase Order
d) Completion Certificate
- Q.8 What is the formula to calculate the number of poles required in LT line distribution?
- a) Length/ Span + 1 b) Length/ Span + 10
c) Span/ Length + 1 d) Span / Length + 10
- Q.9 The process of anticipating the total cost of an industrial installation project is known as?
- a) Costing b) Estimation
c) Valuation d) Depreciation
- Q.10 In electrical installation costing the term 'Contingencies' refers to?
- a) Costs of materials
b) Unexpected or incidental expenses
c) Labor charges
d) Contractor profit

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Tender is an _____.
Q.12 Isolator always operated at _____ load.
Q.13 Write full form of DC.
Q.14 Define line support.
Q.15 3 phase means _____.
Q.16 Which material is most commonly used for conduits in modern concealed wiring?
Q.17 Transformer is an _____ device.
Q.18 Write full form of ELCB.
Q.19 Cleat wiring is suitable for which of installation?
Q.20 HT lines mean _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 List any three selection criteria of cables.
Q.22 Explain earthing of an electrical installation.
Q.23 With proper justification required find the number of sub circuits.
Q.24 Define labour charges.
Q.25 Write the steps for estimating and costing of industrial wiring.
Q.26 Write short note on IS and IE rules.

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